

Total No. of Questions : 8]

SEAT No. :

PC-1957

[Total No. of Pages :2

[6353]-291

T.E. (Cyber Security Honors Course)
INFORMATION AND CYBER SECURITY
(2019 Pattern) (310401) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) Explain in detail Diffie-Hellman. Discuss the vulnerability of the Diffie-Heilman key exchange algorithm. **[8]**

- b) Explain the key steps involved in solving a system of modular congruences using the Chinese Remainder Theorem. How does the choice of pairwise coprime moduli contribute to its effectiveness? **[9]**

OR

Q2) a) Describe the role of the public key and private key in the RSA algorithm. How do they work together to enable secure communication? **[8]**

- b) Explore the advantages of Elliptic Curve Cryptography over traditional public-key cryptosystems, such as RSA. How does ECC provide equivalent security with shorter key lengths? **[9]**

Q3) a) What is cybercrime and what are different types of cybercrimes explain in detail with examples? **[9]**

- b) Explain Social engineering with example? Explain social effects of Cyber Stalking? **[9]**

OR

Q4) a) Outline the procedural steps involved in identifying and assessing risks as part of the information security risk management process? **[9]**

- b) Categorize the various types of laws, and provide a clear distinction between the concepts of law and ethics? **[9]**

P.T.O.

Q5) a) Illustrate the components of SSL with a clear diagram, and elucidate the distinctions between an SSL connection and an SSL session? **[8]**

b) What functionalities does IPSec offer, and what are the contrasting characteristics between Transport mode and Tunnel mode within IPSec? **[9]**

OR

Q6) a) What is the role of X.509 in cryptography? How does X.509 contribute to the field of cryptography and what role does it play in ensuring secure communication? **[8]**

b) What is email security and why it is necessary? Explain any one algorithm used for email security **[9]**

Q7) a) Explain phishing attack, including an illustrative example and outline the various types of phishing strategies? **[9]**

b) Define computer worms and viruses, explain the mechanisms through which computer viruses spread, and outline effective strategies for safeguarding against these cybersecurity threats? **[9]**

OR

Q8) a) Provide a brief overview of Intrusion Detection Systems (IDS), covering their types, along with a discussion on the limitations and challenges associated with these systems? **[9]**

b) What is Malware? Enlist different types of malware. What precaution needs to protect from malware. **[9]**

